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Redefining service quality: Development and validation of the robotic service quality scale

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ABSTRACT

The expansion of robotic technologies in the hospitality industry is redefining traditional service interactions and customer expectations. Grounded in social exchange theory, this study presents a multidimensional framework for measuring the quality of robotic services in hospitality settings. Through a comprehensive literature review, five focus group studies, and a four-stage scale development process involving 1600 valid responses, this study develops and validates a multidimensional framework to assess hotel customers' perceptions of robotic service quality (ROBOQUAL). The resulting framework comprises three key dimensions: robotic service quality, robotic task efficiency, and robotic empathy. The study highlights the transformative role of robotic service quality in shaping hospitality experiences, setting the foundation for future research in AI-driven service innovation. Theoretically, the validation of the ROBOQUAL scale provides a robust tool for future research by expanding the social exchange theory into the context of human-robot interactions (HRI). Practically, the insights gained from this study can guide hospitality managers in integrating robotic services more effectively to enhance customer satisfaction and operational efficiency.

1. Introduction

The intersection of artificial intelligence (AI), robotics, and hospitality has introduced transformative changes in how services are conceptualized, delivered, and evaluated (Saputra et al., 2024; Soliman et al., 2023). Over the past two decades, hotels and related businesses have evolved from offering primarily human-driven interactions to exploring the potential of automating customer-facing processes (Naumov, 2019). Strides in computing power, sensor technologies, data processing capabilities, and machine-learning algorithms have spurred these advancements (Makridakis, 2017; Wirtz et al., 2021). As a result, it is increasingly feasible to deploy robots that can autonomously handle tasks once reserved for humans, from customer check-in to luggage handling, room service delivery, and concierge services (Murphy et al., 2019; Xu et al., 2023). Hotels that successfully adopt robots aim to capture multiple benefits, including operational efficiency, consistency in service delivery, and the capacity to gather and process customer data

in real-time (Prentice and Nguyen, 2021). This data-centric approach allows for customized service offerings and dynamic adjustments (Kim et al., 2022). Importantly, automation in hospitality spans multiple forms of robotics, AI, and service automation, including physically embodied service robots as well as kiosk-like self-service technologies. By occupying physical space and acting as social entities rather than just transactional interfaces, embodied robots create a distinct set of expectations for the customer experience.

In the hospitality context, service quality has traditionally relied on human interactions and intangible cues that shape customer perceptions (Lin et al., 2020). As robots enter these contexts, many industry stakeholders question whether automated solutions can deliver an equivalent or superior level of care, empathy, and personalized attention (Tung and Au, 2018). The concept of "robotic service quality" emerges to define how customers evaluate robots in roles previously fulfilled by human staff (Song et al., 2023). This need becomes clearer when contrasted with foundational human-centered service quality measures.

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SERVQUAL, for example, is rooted in the expectations–perceptions (gap) logic and operationalizes service quality through dimensions such as tangibles, reliability, responsiveness, assurance, and empathy–dimensions originally specified for employee-delivered service encounters (Parasuraman et al., 1985, 1988).

Despite promising developments, hospitality research has lacked a dedicated and systematically validated tool for capturing customer perceptions of robotic service quality (Murphy et al., 2019; Prentice and Nguyen, 2021). Existing measurement tools often adapt or extend measures designed initially for human service encounters, e.g., SERVQUAL (Parasuraman et al., 1985, 1988), and LODGSERV (Knutson et al., 1990), ignoring the unique attributes that differentiate robots from human staff. While SERVQUAL's dimensions (reliability, assurance, tangibles, empathy, and responsiveness) effectively capture human-to-human rapport, they fail in robotic contexts because they presume human agency. For instance, "assurance" in SERVQUAL is rooted in human knowledge and courtesy; however, for a robot, assurance is derived from "perceived intelligence" and technical reliability (Kim et al., 2022). Similarly, "empathy" in traditional scales refers to individualized human attention, whereas in robotics, empathy is a manifestation of social-emotional cues and anthropomorphism (Breazeal and Foerst, 1999; Chi et al., 2024). Humans possess innate empathy and the capacity for nuanced conversations, while robots rely on algorithmic decision-making and sensor-driven responses (Paluch et al., 2020; Wirtz et al., 2021). These fundamental differences mean that traditional scales overlook the "uniqueness neglect" that often occurs when customers interact with non-human agents (Kim et al., 2022), necessitating a fresh perspective and a novel measurement tool.

In the hospitality and tourism context, various studies explored how service technologies influence customer satisfaction, behavioral intentions, and operational efficiency (Bowen and Morosan, 2018; Kim and Qu, 2014). Many have relied on established technology acceptance models, such as the Technology Acceptance Model (TAM) (Davis et al., 1989) and the Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003). However, these models often overlook how operational and relational dynamics shift when robots are integrated into service environments (Naumov, 2019). ROBOQUAL differs from these utility-focused models by incorporating the social exchange elements inherent in hotel services. While models like the Service Robot Integration Willingness (RIWS) scale (Lu et al., 2019) focus on the long-term predisposition of a customer to accept a robot, and the Social Service Robot Interaction Trust (SSRIT) scale (Chi et al., 2021) focuses specifically on trust toward AI agents, ROBOQUAL evaluates the immediate quality of the service encounter itself. Furthermore, previous scales measuring anthropomorphism (SSRA) focus on the robot's "personhood" (Chi et al., 2024), whereas ROBOQUAL synthesizes these social cues into a framework of service delivery quality, including task efficiency and simulated empathy.

Anthropomorphized robots with advanced speech recognition may yield different customer perceptions than simpler kiosk-like robots focused on one or two tasks (Belanche et al., 2021). Moreover, the intangible aspects of empathy, warmth, and psychological comfort, which have long been emphasized in the hospitality context, are also relevant to robotic service encounters but may manifest differently (Chuah et al., 2021). Studies that have proposed scales related to robots often do so in generalized technology acceptance contexts, lacking the granularity needed for hospitality (Liu et al., 2024). Many rely on broad constructs or the repurposing of instruments for self-service technologies. While valuable, these approaches fail to fully capture the intricacies of how customers interact with and evaluate physically present, autonomously moving, human-like, or semi-humanoid service robots, which represent a distinct "social entity" rather than a mere tool (Gursoy et al., 2019; van Doorn et al., 2016). A comprehensive scale must account for mechanical competence, anthropomorphism, perceived intelligence, environmental synergy, and emotional or empathic capacities (Prentice and Nguyen, 2021; Song et al., 2023).

Building on these insights, further research is required to refine our understanding of robotics in the hospitality context, particularly by addressing the limitations of existing measures. Responding to Pijls et al. (2017) and Murphy et al. (2019), this study seeks to develop a robust instrument to measure hotel customers' perceptions of robotic service quality (hereafter ROBOQUAL), an area that remains underexplored. By recognizing this gap, the present study seeks to establish a foundational framework for examining how robotic services shape and influence hotel customers' evaluations of service quality. To achieve this goal, it develops and validates a multi-item ROBOQUAL scale, employing the scale development procedures recommended by Churchill (1979), DeVellis (2016), and Lu et al. (2019).

Following a comprehensive review of the literature, five focus group studies, and a four-stage scale development process, the study conceptualizes and validates a customer-centered model that reflects how hotel customers evaluate service interactions with service robots. Based on 1600 valid responses across three studies, the present study developed and refined an 11-item, three-dimensional ROBOQUAL scale, which consists of robotic service quality, task efficiency, and empathy. As robots become more prevalent in hospitality settings, an opportunity emerges to explore how these emerging technologies redefine traditional service paradigms, transform customer expectations, and reinforce the critical link between the conceptual investigation of service robots' impact on the customer experience and empirical assessment via the ROBOQUAL scale.

2. Theoretical background

2.1. Robotics in the hospitality industry

A robot can be defined as an actuated mechanism equipped with multiple axes and varying degrees of autonomy to perform tasks in its environment (ISO, 2012). Their contemporary form emerged in the late twentieth century following advancements in computer science and cybernetics, which granted robots increased intelligence and autonomy (Raj and Kos, 2022). Robotics, an interdisciplinary field integrating engineering, computer science, and information technology, has enabled machines to undertake tasks that are hazardous or impractical for humans (Soliman et al., 2023).

In recent decades, the range of robotic applications has expanded beyond industrial manufacturing (Pagliarini and Lund, 2017) into healthcare (Silvera-Tawil, 2024), space exploration (Herpers, 2025), and personal assistance (Kipnis et al., 2022). Technological advances, including enhanced computational power, miniaturization of electronics, and improved materials, have led to more capable and intelligent robotic systems (Ye et al., 2022). These developments have also permeated the hospitality industry, where robots support front-office operations and guest services, thereby reshaping the customer experience (Ivanov and Webster, 2017). Major hotel chains such as Aloft Hotels, InterContinental Hotels Group, and Marriott properties have embraced these innovations, reflecting a growing trend in robot-assisted hospitality operations (Pizam et al., 2022). The literature suggests that customers are more likely to interact with humanoid robots (Saputra et al., 2024), which contributes to new forms of service interactions and shifts in service expectations.

Research on service robots in the existing hospitality research can be categorized into four theoretical domains. First, a robot-centered approach investigates attributes such as anthropomorphism, efficiency, and perceived trust (So et al., 2024; Song et al., 2024). Second, a human resources dimension focuses on employee perceptions, value co-creation, talent management, and job replacement (Brougham and Haar, 2018; Wang et al., 2024). Third, a model-building school develops conceptual frameworks to explore adoption and implementation (Go et al., 2020; Kang et al., 2022). Fourth, robots as service agents highlight how service quality and the service environment evolve with robotic integration, framing service as a multidimensional product subject to

expectations and performance evaluations (Lu et al., 2021; Zhang et al., 2022). These four domains highlight the hospitality industry's growing adoption of robotic technologies to improve both customer experiences and operational efficiency.

2.2. Robotic service quality

The phenomenon of robotic service quality builds upon traditional service quality frameworks while addressing the unique characteristics of automated and AI-driven service encounters. As artificially intelligent service robots increasingly mediate customer interactions in hospitality settings, it becomes essential to evaluate their performance, responsiveness, and reliability in shaping service experiences (Belanche et al., 2020). Drawing upon Murphy, Gretzel, and Pesonen's (2019) robotic service concept, which defines the service robot as the service delivery channel, and adopting Kim, So, and Wirtz's (2022) reciprocal exchange approach, the present study defines robotic service quality as the service robot's level of efficiency, expediency, and adaptability perceived by the customer during the service interaction. Since robotic services differ from human interactions, customers' evaluations of technical precision, consistency, and error recovery mechanisms significantly shape their perceptions of service quality (Tussyadiah and Park, 2017).

In technology-enabled service encounters, customers evaluate the service agent using fundamental social-cognition mechanisms, relying primarily on inferences about competence and warmth (Fiske et al., 2007; Cuddy et al., 2011), and this logic has been applied to customer evaluations of technology-based and robotic service agents (van Doorn et al., 2016; Belanche et al., 2021). Prior HRI research posits that these warmth and competence inferences are effective in capturing how customers perceive automated social presence (ASP) and in explaining downstream service outcomes in technology-infused service settings (van Doorn et al., 2016).

Accordingly, robotic service quality is treated as a structured evaluation in which customers form an overall judgment informed by more specific functional and socio-emotional assessments. Building on the work of Wirtz et al. (2018) and Belanche et al. (2021), robotic service quality is conceptualized not as a set of interchangeable indicators, but

as a hierarchical synthesis of evaluative judgments. Specifically, a distinction is made between robotic service quality (a holistic, high-level evaluative judgment of the encounter) and task efficiency and empathy (framed as foundational, domain-specific evaluative components). This distinction is consistent with the service robot acceptance model (sRAM), which suggests that customers form global perceptions based on a combination of functional performance and social-emotional cues (Wirtz et al., 2018). Furthermore, the conceptualization of these dimensions is grounded in social cognition research, which identifies competence and warmth as the two universal pillars of human impression formation (Cuddy et al., 2011; Fiske et al., 2007). Because service robots are increasingly perceived by customers as socially present actors rather than mere tools, evaluations of their service quality naturally align with these core social dimensions (van Doorn et al., 2016; Belanche et al., 2021). Within this framework, task efficiency and empathy are not viewed as interchangeable indicators, but as conceptually distinct facets reflecting the robot's functional competence and socio-emotional warmth, respectively. While task efficiency and empathy represent the granular "sense-think-act" capabilities of the robot, the robotic service quality dimension captures the customer's summary assessment of the overall interaction. Thus, these dimensions are viewed as "different in nature" because they represent different stages of the cognitive evaluation process: one being a summative holistic perception and the others being the specific performance drivers that inform that perception (Belanche et al., 2021; Wirtz et al., 2018).

Prior studies operationalized customers' robot- and AI-enabled service perceptions using a range of instruments that vary in construct focus and domain specificity (see Table 1). These include scales labeled as robotic service quality (Prentice and Nguyen, 2021), as well as closely related measures that capture adjacent constructs such as social service robot interaction trust (SSRiT; Chi et al., 2021), service robot integration willingness (SRIW; Lu et al., 2019), social robot anthropomorphism (SSRA; Chi et al., 2024), social-robot service quality developed adapting SERVQUAL (SERVBOT; Kharub et al., 2021), and AI service agents service quality (AISAQUAL; Noor et al., 2022). Taken together, this body of work indicates that (i) most existing instruments are developed in general service contexts rather than hospitality-specific settings, (ii)

Table 1
Overview of Existing Service Robot Measurement Scales.

Scale	Author(s)	Context	Dimensions	Key limitations
Robotic Service Quality	Prentice and Nguyen (2021)	General services	1. Automization 2. Personalization 3. Precision 4. Efficiency	- Not hospitality specific - Developed from existing scales - Focus on the functional and technical performance of service robots
Social Service Robot Interaction Trust (SSRiT)	Chi et al. (2021)	General services	1. Trustworthy robot function and design 2. Propensity to trust in a robot 3. Trustworthy service task and context	- Not hospitality specific - Measures trust, not robotic service quality
Service Robot Integration Willingness (SRIW)	Lu et al. (2019)	General services	1. Performance efficacy 2. Intrinsic motivation 3. Anthropomorphism 4. Social influence 5. Facilitating condition 6. Emotions	- Not hospitality specific - Measures willingness to integrate service robots, not robotic service quality
Scale of Social Robot Anthropomorphism (SSRA)	Chi et al. (2024)	General services	1. Human-like appearance 2. Social intelligence 3. Emotional capacity 4. Self-understanding	- Not hospitality specific - Measures robot anthropomorphism, not robotic service quality
Social Robot Service Quality (SERVBOT)	Kharub et al. (2021)	General services	1. Reliability 2. Responsiveness 3. Assurance Empathy 4. Entertainment	- Not hospitality specific - Does not measure robotic service quality - Developed from existing scales
Artificial Intelligence Service Agents Service Quality (AISAQUAL)	Noor et al. (2022)	General services	1. Efficiency 2. Security 3. Availability 4. Enjoyment 5. Contact 6. Anthropomorphism	- Not hospitality specific - Agent (chatbots and virtual assistant) focus - Developed from existing scales

several measures assess constructs related to robot-enabled services, e. g., trust, integration willingness, anthropomorphism, rather than robotic service quality, and (iii) even when measuring "service quality", the scale development processes relied heavily on the existing scales. Table 1 provides a comparative summary of these existing instruments, detailing their specific dimensions, development contexts, and the gaps that necessitate a new approach.

Despite the aforementioned adaptations and various applications of SERVQUAL to enhance its effectiveness, a reliable and replicable scale for measuring robotic service quality in hospitality has yet to be established. To the best of the authors' knowledge, the sole scale developed to assess customers' perceptions of robotic service quality, proposed by Prentice and Nguyen (2021), was not tailored to a specific service setting and was adapted from existing scales. Therefore, this study develops and validates a new multi-item ROBOQUAL scale to address this gap, using scale development procedures recommended by Churchill (1979), DeVellis (2016), and Netemeyer et al. (2003). In doing so, ROBOQUAL is positioned to extend the existing literature by providing a hospitality-specific instrument tailored to hotel customers' perceptions of robotic service quality.

Developing a robust instrument to measure hotel customers' perceptions of robotic service quality not only fills this measurement gap but also provides a framework for understanding its broader implications for customer experience. Evaluating the services provided by artificially intelligent service agents has been increasingly linked to customers' experience quality perceptions (Naumov, 2019; Rana et al., 2025). High-quality robotic service fosters both cognitive and emotional engagement, influencing overall satisfaction and reinforcing positive hospitality experiences (Borghi et al., 2023). Customers who perceive seamless, reliable, and intuitive robotic service are more likely to assess their experience positively, increasing trust and long-term adoption of AI-driven hospitality solutions (Lin and Lee, 2025). In light of the expanding role of robotic technologies, this study develops and validates a foundational framework for assessing their service quality, informing both scholarly discourse and strategic decision-making in AI-enabled hospitality environments.

2.3. Social exchange theory

Social exchange theory offers a foundational lens for understanding interactions between customers and service providers, whether human or robotic, by conceptualizing relationships as reciprocal exchanges driven by perceived benefits and costs (Blau, 1964; Homans, 1958). In the context of robotic service delivery, this framework helps explain how customers evaluate interactions with service robots based on expectations of efficiency, reliability, responsiveness, and empathy. Social exchange theory posits that individuals form and maintain social relationships when they perceive the outcomes to be rewarding relative to the investment (Emerson, 1976; Thibaut and Kelley, 1967). Applied to hospitality, this implies that customers are more likely to engage positively with service robots when they perceive high-quality service attributes, such as prompt task execution or emotionally responsive behavior, as outweighing the novelty or discomfort associated with non-human interactions.

Jiang and Kim (2015a) explored social exchange theory in the green hotel context, demonstrating that customers are more likely to support sustainable initiatives when they perceive clear benefits. This concept similarly informs how customers perceive robotic services. Ivanov and Webster (2017) emphasized that the adoption of service automation in hospitality is shaped by perceived mutual value, suggesting that social exchange theory can explain the balance customers seek between convenience and emotional connection in robotic service experiences. Kim et al. (2022) demonstrated that customers' perceptions of fairness, reciprocity, and responsiveness in HRI significantly influence their willingness to accept and engage with robotic services. Similarly, Morosan and DeFranco (2021) showed that social exchange theory

could explain customer trust and acceptance of AI-based hotel technologies during periods of service disruption, highlighting the theory's relevance in assessing technological interactions in hospitality. Liu et al. (2023) further support this view by finding that customers' sense of exclusion during service encounters can lead to perceptions of injustice, diminishing trust, and potentially triggering retaliatory behaviors, thus underscoring the importance of perceived social equity in HRI.

Building on this foundation, the present study adopts social exchange theory to conceptualize robotic service encounters as reciprocal exchanges, where customer perceptions of robotic efficiency, task performance, and empathetic response serve as signals of value. This theoretical perspective provides the scaffolding for the development of the ROBOQUAL scale, guiding both the operationalization of robotic service quality and offering a promising basis for future research exploring how these dimensions influence customers' behavioral intentions.

3. Scale development and results

3.1. Research design

This study undertakes the development and validation of a multi-item ROBOQUAL scale, adhering to established scale development procedures as outlined by Churchill (1979), DeVellis (2016), and Lu et al. (2019). Grounded in an extensive review of the literature, the research follows a rigorous four-stage scale development process (see Fig. 1).

3.1.1. Item generation

The implementation of the scale development unfolded in four distinct phases, as presented in Fig. 1. Importantly, the core dimensions of the scale, specifically robotic service quality, task efficiency, and empathy, were not selected *a priori* or prioritized arbitrarily at the outset of the study. Instead, they emerged as the most conceptually and statistically robust factors through the systematic, multi-stage empirical procedures summarized in Fig. 1. The item generation phase (Stage 1) comprised an in-depth review of the current literature concerning service quality, service robots, HRI, and AI in the hospitality and tourism contexts to identify the domain and establish the conceptual definitions. This phase was followed by conducting five focus group studies with 40 participants recruited from a higher education institution in the South-eastern United States. The eligibility criteria required participants to be at least 18 years old and reside in the U.S. Throughout the focus group sessions, participants were engaged with scenarios and queries that progressed from general discussions about artificially intelligent service robots to more detailed, realistic situations. Responses were recorded using a voice recorder and later transcribed using Microsoft 365. Informed by prior research, e.g., Balan et al. (2015), this study opted for manual coding to create an initial pool of items, capitalizing on the researchers' familiarity with the subject matter to accurately capture nuanced meanings and avoid potential errors that automated algorithms might introduce. Using the transcripts, an initial pool of 300 items was created and subsequently reduced by eliminating redundant items to enhance the pool's clarity and efficiency. This process resulted in a pool of 280 items.

In the next step, the revised item pool underwent a content validity assessment, during which seven experts evaluated each item for representativeness, specificity, and clarity, as recommended by Boateng et al. (2018). This panel comprised academics (professors and associate professors) from hospitality and marketing departments at various US universities. Their collective expertise included numerous publications in prominent journals on service quality, technology adoption, and service robots. Informed by expert input, the initial item pool was refined through item revision and elimination to ensure alignment with the conceptual definition of robotic service quality. Items deemed insufficiently representative, specific, or clear by at least two experts

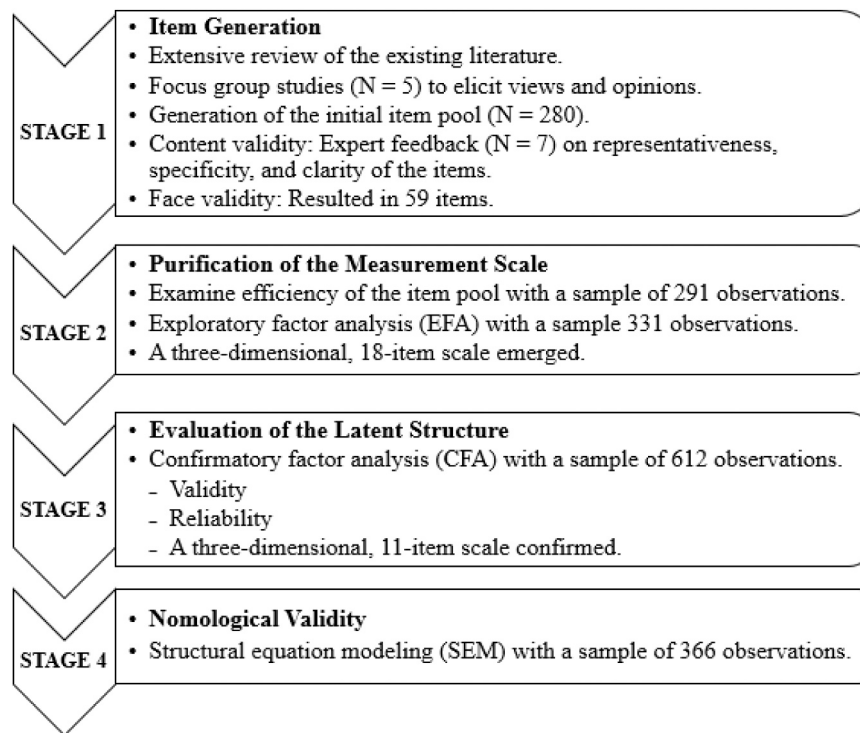


Fig. 1. Scale development process.

were removed, resulting in a more parsimonious pool of 83 items (Lu et al., 2019). Following this stage, face validity was examined to determine whether the items appropriately reflected the construct definition, based solely on the researcher's judgment (Hair et al., 2019). The face validity analysis resulted in 59 items that reflected hotel customers' perceptions of robotic service quality (see Table A2).

These 59 items served as the empirical foundation for the subsequent refinement stages described in the following sections. As the study progressed through pilot-testing and exploratory factor analysis (EFA; N = 331), the pool was reduced to an 18-item, three-factor structure. This was further refined through confirmatory factor analysis (CFA; N = 612) into a final 11-item, three-dimensional solution that was then nomologically validated (see Tables A5 and 2). By following this rigorous, data-driven trajectory, robotic service quality, task efficiency, and empathy were retained because they represent the most significant facets of hotel customers' actual perceptions, ensuring the ROBOQUAL scale's dimensions are grounded in empirical evidence rather than theoretical assumptions.

3.1.2. Scale purification

Scale purification (Stage 2) was conducted using data collected at three intervals from two separate sources. The first sample, comprising 291 valid responses collected via the MTurk platform, was used to pilot-test the item pool (N = 59) that emerged from the item generation and content validity procedures (see Table A3 for demographic profiles). Consistent with recommended scale development practices (Boateng et al., 2018; Lu et al., 2019), the purpose of this pilot testing stage was to assess item functioning and clarity. Based on insights obtained from this stage, revisions to item wording and phrasing were made to enhance clarity and consistency, while the full item pool was retained for subsequent analysis.

A second sample of 331 valid responses was collected from Prolific to refine the item pool and identify its latent structure through an EFA (see Table A4 for demographic profiles). The suitability of EFA was confirmed by Bartlett's Test of Sphericity (χ^2 (153) = 4595, $p < 0.01$), indicating that item correlations were sufficiently robust for factor

analysis (Bartlett, 1937) (see Table A11). The EFA, using the maximum likelihood extraction method and Varimax (orthogonal) rotation, resulted in three meaningful clusters, i.e., factors, comprising 18 items. These three factors with eigenvalues greater than 1 accounted for 52.45 %, 10.94 %, and 6.34 % of the variance, respectively, explaining 69.74 % of the total variance collectively (Fabrigar et al., 1999). This suggests that the three identified factors effectively represent the underlying dimensions within the data. Adhering to the recommended practice, each factor retained at least three items to ensure construct reliability (Bollen, 1989). All retained items exhibited substantial factor loadings (> 0.5), reflecting strong relationships between the items and their respective factors. A total of 41 items were removed from the analysis either due to significant cross-loadings (above 0.5) or weak item-to-factor loadings (below 0.5).

3.1.3. Scale validation

To validate the 18-item, three-factor solution derived from EFA (see Table A12), a CFA was conducted in Stage 3. Building on the EFA results obtained from the second sample, data were collected from 624 participants, with 612 responses remaining after excluding 12 for unsuccessful attention checks and invariant answers (see Table A5 for demographic profiles). This sample was used to further refine the three-factor structure identified via EFA. The dataset was examined for missing data, outliers, and univariate normality. Univariate normality was assessed based on the recommendation that skewness and kurtosis values remain below a threshold of 2 (Demir, 2022). The normal distribution of the dataset was confirmed through visual inspection of histograms and box plots. During CFA, items exhibiting insufficient factor loadings or evidence of multicollinearity were removed. This process culminated in a three-factor, eleven-item solution (see Table 2).

To comprehensively evaluate the latent structure, the measurement model's construct validity was assessed in the next step, focusing specifically on convergent and discriminant validity (Lu et al., 2019). The resulting chi-square statistic was statistically significant ($\chi^2 = 196.09$, $p < 0.001$), and the normed chi-square value ($\chi^2/df = 4.78$) remained below the recommended threshold of 5, signifying an acceptable fit.

Table 2
Validity and reliability of constructs.

Constructs	Items	Loadings	AVE	α	CR
Robotic Service - Quality	1. Artificially intelligent service robots are capable of providing exceptional service.	0.900	0.733	0.947	0.931
	2. Artificially intelligent service robots can provide exceptional service.	0.870			
	3. Artificially intelligent service robots can deliver a positive service experience.	0.840			
	4. Artificially intelligent service robots can meet the expectations of customers who expect quality service.	0.830			
	5. Artificially intelligent service robots can deliver quality service.	0.840			
Robotic Service - Efficiency	1. Artificially intelligent service robots focus solely on tasks, avoiding distractions.	0.760	0.554	0.758	0.788
	2. I think artificially intelligent service robots will not be easily distracted.	0.740			
	3. Artificially intelligent service robots are knowledgeable about the services provided at the hotel.	0.730			
Robotic Service - Empathy	1. Artificially intelligent service robots can sense when a customer is upset.	0.830	0.685	0.905	0.867
	2. Artificially intelligent service robots can understand customers' reactions.	0.880			
	3. Artificially intelligent service robots can create a bond with customers.	0.770			

Multiple fit indices were employed to assess the adequacy of the hypothesized three-factor model, providing insight into how accurately the model reproduces the observed covariance matrix among the measured items (Hair et al., 2019). Both absolute and relative fit indices were employed following the best practices outlined by Kline (2016). Absolute fit indices, including the Tucker-Lewis Index (TLI = 0.95), Normed Fit Index (NFI = 0.96), Adjusted Goodness of Fit Index (AGFI = 0.91), and Goodness of Fit Index (GFI = 0.95), indicated good model fit. Similarly, relative fit indices, i.e., the Comparative Fit Index (CFI = 0.97) and the Root Mean Square Error of Approximation (RMSEA $\leq$ 0.07), further supported the model's adequacy (Hu and Bentler, 1999; Kline, 2016).

In the next step, convergent validity was examined to determine the extent to which measures designed to assess the same construct correlate with each other. High correlations among these measures indicate an effective representation of the intended construct (Fornell and Larcker, 1981). This study utilized the average variance extracted (AVE) to assess convergent validity. Results revealed that the AVE values for all three constructs surpassed the recommended threshold of 0.5, confirming acceptable convergent validity (see Table 2). Following this, discriminant validity was assessed to ensure that theoretically distinct constructs remain empirically separate. Discriminant validity was verified by comparing the AVE values of each construct with their respective squared correlations against other constructs. The findings

demonstrated that the AVE of each construct exceeded its squared correlation with every other construct in the model (see Table 3), thereby establishing discriminant validity (Fornell and Larcker, 1981).

Reliability of the new ROBOQUAL scale was conducted at two distinct levels: (1) item reliability and (2) construct reliability (Fornell and Larcker, 1981). Item reliability in this study was assessed by calculating the squared factor loadings. All eleven items of the proposed scale exceeded the recommended threshold of 0.50 (approximately equivalent to a standardized loading of 0.7), confirming acceptable item reliability (Hair et al., 2017). Construct reliability, which evaluates the accuracy of a measurement instrument in representing an underlying factor, is recommended to exceed a threshold of 0.7 for Cronbach's alpha, a widely utilized measure of internal consistency (Hair et al., 2017). All three constructs demonstrated Cronbach's alpha values greater than 0.7, establishing robust construct reliability (see Table 2).

Consistent with the established scale development procedures, the dimensions derived from the EFA were named based on the content and thematic characteristics of the items (Netemeyer et al., 2003): a) robotic service quality, b) robotic efficiency, and c) robotic empathy.

3.1.4. Model comparison

Following the approach of Brakus et al. (2009), four competing models were evaluated using a sample of 366 valid responses to identify the most appropriate factor structure for the scale. Specifically, Model 1 tested a one-factor solution, with all items loading onto a single latent construct. Model 2 examined a first-order factor model, in which the three dimensions were treated as correlated but distinct factors. Model 3 represented a second-order reflective model, where the three first-order dimensions loaded onto a higher-order latent factor of robotic service quality. Finally, Model 4 tested a second-order formative model, in which the higher-order construct of robotic service quality was formed by its three underlying dimensions (see Table 4). This second-order formative specification is also conceptually appropriate because task efficiency and empathy represent distinct evaluative components that contribute to the higher-order appraisal of robotic service quality, rather than interchangeable indicators at the same level of abstraction. In other words, customers may weigh functional gains (e.g., speed, accuracy) and socio-emotional performance (e.g., warmth, understanding) when arriving at an overall service-quality judgment in robotic encounters.

The comparison of these competing models indicated that the construct is best represented as a multidimensional, higher-order structure. Although the first-order factor model yielded the lowest AIC value (833), its overall fit indices (CFI = 0.77, RMSEA = 0.22, SRMR = 0.20) reflected a poor model fit. In contrast, the second-order formative model (Model 4) achieved substantially better fit statistics (χ^2/df = 3.96, CFI = 0.93, RMSEA = 0.09, SRMR = 0.06), demonstrating both statistical adequacy and conceptual soundness (Hu and Bentler, 1999; Kline, 2016). Accordingly, robotic service quality is best conceptualized as a higher-order formative construct composed of three interrelated dimensions, reflecting a coherent and parsimonious representation of the underlying measurement structure.

3.1.5. Nomological validity

To establish nomological validity, this study tested the relationships between the newly developed ROBOQUAL scale and two existing constructs within the domain of service quality, namely WOM intentions

Table 3
Fornell-Larcker Criterion.

Constructs	2	3	4
Robotic Service - Efficiency	0.820		
Robotic Service - Empathy	0.231	0.917	
Robotic Service - Quality	0.585	0.481	0.909

Values on the diagonal (bolded) are the square root of the average variance extracted (AVE), while the off-diagonals are correlations.

Table 4
CFA model fit comparison.

Model	Description	χ^2	df	χ^2/df	p	CFI	RMSEA	SRMR	AIC
1	One factor	970	44	22.05	0.00	0.71	0.24	0.14	1014
2	First-order factor	785	42	18.69	0.00	0.77	0.22	0.20	833
3	Second-order factor (reflective)	958	225	4.26	0.00	0.92	0.09	0.08	1060
4	Second-order factor (formative)	879	222	3.96	0.00	0.93	0.09	0.06	987

(Alexandrov et al., 2013; Harrison-Walker, 2001) and satisfaction (Roggeveen et al., 2007; Tusssyadiah, 2016). Minor adjustments were made to the measurement items to ensure contextual alignment and to enhance measurement validity. All constructs were measured using a seven-point Likert-type scale, ranging from 1 (“Strongly Disagree”) to 7 (“Strongly Agree”) (see Tables A1 and A7). 422 responses were initially collected. Data screening procedures involved removing incomplete responses, examining missing data patterns to ensure the absence of systematic bias, and assessing univariate normality. This process yielded a sample of 366, which was retained for the statistical analysis (see Table A6 for demographic profiles). Collinearity diagnostics indicated that variance inflation factor (VIF) values for the indicators ranged from 1.31 to 1.88, remaining below the recommended threshold of 5 (Hair et al., 2017). A common method bias assessment was also conducted using Harman’s single-factor test (Podsakoff et al., 2003). The results of the unrotated factor analysis indicated that the first factor accounted for 26.21 % of the total variance, which is below the recommended 50 % threshold, suggesting that common method bias is unlikely to be a significant concern in this study.

The proposed relationships were tested utilizing Partial Least Squares Structural Equation Modeling (PLS-SEM) via SmartPLS Version 4.1.1.1 (Ringle et al., 2014). Following the two-stage process recommended by Hair et al. (2017), the study initially assessed the measurement model to confirm reliability and validity and subsequently examined the structural model. Bootstrapping procedures with 5000 resamples were used to evaluate the significance of both path coefficients and factor loadings.

3.1.5.1. Measurement model. Once the normality of the dataset was established through visual inspection of histograms and box plots, composite reliability (CR) estimates were assessed to evaluate the reliability of the measurement scales (Hair et al., 2019). The CR values for all constructs exceeded the 0.7 threshold, providing evidence for measurement scales’ internal consistency and reliability (see Table A7). The construct validity was assessed through the analysis of both convergent and discriminant validity of the measured constructs. Each construct in the proposed model exhibits an AVE value above 0.5, supporting convergent validity (Hair et al., 2019). To assess discriminant validity, two methods were utilized. First, the square root of the AVE for each construct was evaluated against the squared correlations between that construct and all others (see Table A8). Discriminant validity was established as the AVE values were greater than the corresponding squared correlations with other constructs (Fornell and Larcker, 1981). To address the limitations of this approach, Henseler et al. (2015) proposed the heterotrait–monotrait ratio (HTMT) of correlations as a more robust approach to assessing discriminant validity. All HTMT values in the present analysis were below the 0.90 threshold, further confirming discriminant validity (see Table 5).

Table 5
Heterotrait–Monotrait Ratio (HTMT) of Correlations.

Constructs	1	2	3	4
WOM Intentions				
Robotic Efficiency	0.509			
Robotic Empathy	0.432	0.268		
Robotic Service Quality	0.590	0.679	0.520	
Satisfaction	0.872	0.636	0.413	0.684

3.1.5.2. Structural model. To examine the significance of the factor loadings and path coefficients, PLS-SEM algorithms were implemented. Bootstrapping (5000 resamples) was conducted to determine the statistical significance of the hypothesized relationship between the constructs in the model (Chin et al., 2008). The analysis began by assessing the structural relationships among the constructs. Robotic service quality ($\beta = 0.45$; $p < 0.01$), robotic efficiency ($\beta = 0.26$; $p = 0.005$), and robotic empathy ($\beta = 0.11$; $p < 0.001$) each had a significant positive effect on customer satisfaction. Likewise, robotic service quality ($\beta = 0.38$; $p < 0.01$), task efficiency ($\beta = 0.19$; $p < 0.001$), and robotic empathy ($\beta = 0.17$; $p < 0.01$) significantly influenced customers’ WOM intentions (see Fig. 2). These findings confirm the proposed nomological links between the ROBOQUAL dimensions and both outcome variables. The model explained 88.3 % of the variance in satisfaction ($R^2 = 0.883$) and 76.3 % of the variance in WOM intentions ($R^2 = 0.763$). Both R^2 values exceed the 0.26 threshold suggested by Cohen (1988) for substantial explanatory power in structural models. These results not only provided empirical support for the predictive validity of the ROBOQUAL scale but also reinforced its theoretical relevance in explaining customers’ behavioral intentions in the robotic service context (see Table 6).

3.1.6. Scenario method

Originating from Kahn and Wiener’s (1967) systematic model-building approach of the 1960s, the scenario method involves constructing vivid, detailed narratives grounded in empirical observations while incorporating speculative and imaginative elements to explore potential future developments. These carefully crafted scenarios are beneficial in research contexts, as they present realistic service encounters that prompt responses, thus uncovering more profound insights into customers’ perceptions and attitudes toward service quality and HRI. These narratives can represent various hotel service contexts, such as front desk operations, dining experiences, and room service interactions. For instance, a front desk scenario might depict a guest checking in with the assistance of a robotic receptionist, highlighting the robot’s conversational skills and responsiveness to customer inquiries. To ensure that participants consistently perceived the service agents as embodied, humanoid robots rather than disembodied technologies (e.g., kiosks), each scenario was introduced with an AI-generated visual depiction of a distinctly humanoid service robot tailored to the specific service context. These images emphasized anthropomorphic features, physical embodiment, and interactive capabilities, thereby reinforcing the intended conceptualization of the robot across scenarios.

The introductory section of the survey provided participants with a foundational understanding of service robots, establishing common ground and effectively preparing them for subsequent engagement with scenario-based survey questions (see Appendix A for the scenarios and visual stimuli).

4. Discussion and conclusions

4.1. Discussion of results

4.1.1. Scale development

The present study introduces a novel instrument for assessing hotel customers’ perceptions of robotic service quality. It establishes its nomological validity by aligning it with established constructs from the

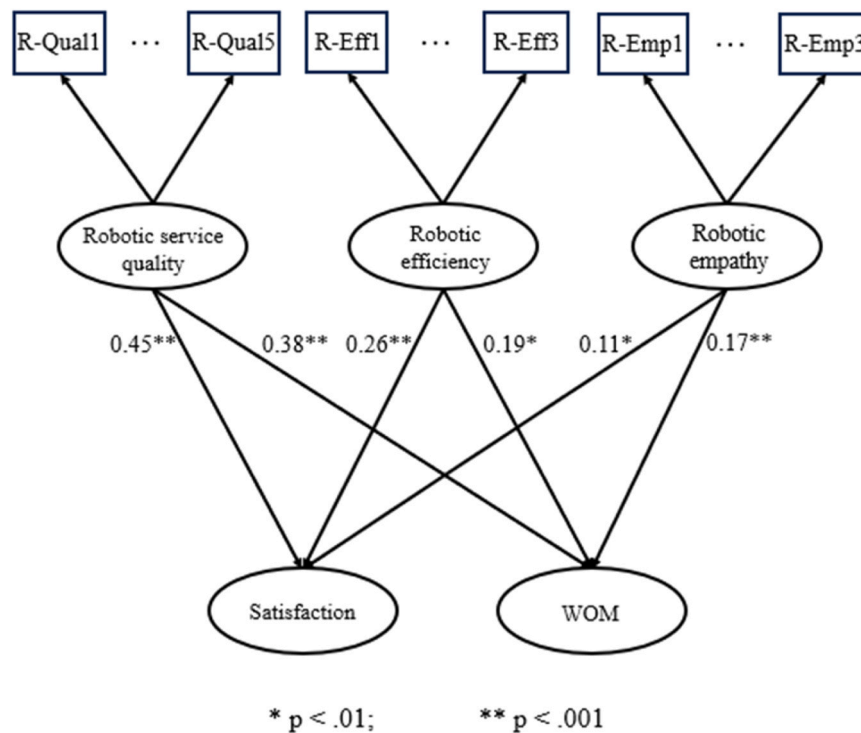


Fig. 2. Results of PLS-SEM.

Table 6
Structural Estimates.

Hypotheses	Path Coefficients	SD	T statistic	Confirmed
H1a: Robotic Service Quality → Satisfaction	0.450	0.064	7.049	Yes
H1b: Robotic Service - Efficiency → Satisfaction	0.261	0.054	4.870	Yes
H1c: Robotic Service - Empathy → Satisfaction	0.111	0.040	2.780	Yes
H2a: Robotic Service Quality → WOM Intentions	0.379	0.072	5.290	Yes
H2b: Robotic Service - Efficiency → WOM Intentions	0.192	0.052	3.677	Yes
H2c: Robotic Service - Empathy → WOM Intentions	0.173	0.049	3.496	Yes

existing literature. The resulting scale, ROBOQUAL, comprises three distinct dimensions: robotic service quality, robotic task efficiency, and robotic empathy, each capturing a different facet of how customers evaluate artificially intelligent service robots in a hotel setting (see Table 2). Specifically, the robotic service quality dimension captures customers' holistic, overall evaluative judgment of the robot's service performance during the encounter, whereas task efficiency and empathy reflect functional and socio-emotional components that inform this judgment. This hierarchical distinction aligns with work on service robots, suggesting that customer evaluations are shaped jointly by functional performance cues (e.g., competence, reliability) and socio-emotional interaction cues (e.g., warmth, rapport), which together inform perceived service quality in technology-mediated encounters (Belanche et al., 2020; Wirtz et al., 2018).

Building on this conceptual ordering, the discussion first elaborates the overarching evaluative dimension before turning to the more specific components. The first dimension, robotic service quality, captures customers' holistic evaluative judgment of the services rendered by a service robot. It is operationalized through five items that evaluate the robots' ability to deliver outstanding service experiences and to meet or

exceed guests' quality expectations (see Table 2). Empirical evidence shows that customers not only recognize but also appreciate robots' capacity to provide superior service (Borghi and Mariani, 2023; Song et al., 2023). Such positive perceptions are pivotal, as they directly influence customer satisfaction and the broader service experience. Rather than reflecting a narrow operational attribute, this dimension represents an overarching assessment that integrates customers' perceptions of multiple service cues encountered during the interaction. By consistently fulfilling or surpassing expectations, service robots enhance both the perceived reliability and the value proposition of a hotel's offerings (Zhong et al., 2021).

The second dimension, robotic service efficiency, measured by three items, captures a service robot's capacity to execute its functions with precision and sustained concentration (see Table 2). By emphasizing uninterrupted task performance and expert knowledge, task efficiency underscores how effectively robots contribute to hotel operations. Customers place a premium on robots that complete tasks both competently and expediently, thereby bolstering overall operational effectiveness (Belanche et al., 2020; Sun et al., 2025). Consistent with the empirical findings, task efficiency emerged as a comparatively stronger predictor of customer outcomes, suggesting that functional competence constitutes a central evaluative criterion in robotic service encounters. Furthermore, when robots reliably handle routine duties, human staff can redirect their efforts toward more complex or personalized customer interactions, optimizing the entire service delivery process (Larivière et al., 2024).

The third dimension, robotic empathy, comprising three items that evaluate the robot's capacity for empathy, measure the ability of service robots to understand and respond to customer emotions (see Table 2). Empathy serves as a key attribute that sets service robots apart from basic automated systems. While the empathy dimension showed weaker path coefficients in the structural model, its statistical significance remained intact. This indicates that socioemotional responsiveness makes a meaningful contribution to customer satisfaction and WOM intentions, though its impact is less pronounced than other factors. This pattern suggests that, in contemporary hotel contexts, customers may

prioritize functional performance over emotional engagement when evaluating service robots. Given the current technological limitations of robotic emotional intelligence, customers may calibrate their expectations accordingly, assigning greater weight to reliability and efficiency than to empathic responsiveness. Nevertheless, by recognizing and responding appropriately to emotional cues, robots retain the potential to enhance customer experiences and foster more individualized interactions, particularly as emotional recognition technologies continue to evolve (Park and Whang, 2022).

An important consideration in interpreting the ROBOQUAL dimensions is that they primarily capture functional aspects of robotic service quality rather than physical design characteristics. While the literature extensively discusses how physical attributes such as appearance, size, and anthropomorphism influence customer perceptions of service robots (Belanche et al., 2021; Saputra et al., 2024), these characteristics did not emerge as distinct dimensions during the scale development. The findings suggest that physical attributes may function as antecedents to service quality perceptions rather than dimensions of service quality itself. That is, a robot's physical design (e.g., humanoid versus mechanoid appearance, facial features, size) may shape initial expectations and acceptance, which subsequently influence how customers evaluate the robot's functional performance across our three dimensions of service quality, task efficiency, and empathy. This interpretation aligns with recent research suggesting that while physical characteristics drive initial reactions and technology acceptance, ongoing quality evaluations depend primarily on functional performance and interaction outcomes (So et al., 2024). For instance, an anthropomorphic robot may generate higher initial empathy expectations. However, customers ultimately evaluate empathy based on the robot's actual ability to sense emotions and respond appropriately during service encounters.

To confirm the multidimensional structure of the ROBOQUAL scale, a series of confirmatory factor analyses (CFAs) were conducted to compare alternative model specifications. Specifically, four competing models were tested: (1) a one-factor model with all items loading onto a single latent construct, (2) a first-order factor model treating the three dimensions as correlated but distinct factors, (3) a second-order reflective model, and (4) a second-order formative model in which the higher-order construct of robotic service quality was formed by its three underlying dimensions. The comparison results indicated that the construct is best represented as a multidimensional, higher-order structure. Although the first-order factor model achieved the lowest AIC value (833), its fit indices (CFI = 0.77, RMSEA = 0.22, SRMR = 0.20) reflected inadequate model fit. In contrast, the second-order formative model (Model 4) demonstrated substantially stronger overall fit ($\chi^2/df = 3.96$, CFI = 0.93, RMSEA = 0.09, SRMR = 0.06) and superior theoretical coherence. Collectively, these findings suggest that robotic service quality functions as a higher-order formative construct composed of three interrelated yet conceptually distinct dimensions, providing both statistical robustness and theoretical consistency.

4.2. Implications

4.2.1. Theoretical implications

Central to this study is the integration of artificially intelligent agents, such as service robots, into hospitality contexts, with a particular focus on examining hotel customers' perceptions of robotic service quality. The growing implementation of service robots across various hospitality settings marks a transformative shift from traditional high-touch, low-tech interactions toward sophisticated, technology-driven service delivery systems. Introducing ROBOQUAL, a multidimensional instrument specifically designed to evaluate hotel customers' perceptions of robotic service quality, represents a pioneering initiative in quantifying the influence of robotic interactions on customer perceptions. It addresses a notable gap within the existing literature, where robotic service quality has received limited attention as an antecedent of

hotel customers' behavioral intentions. Drawing on social exchange theory (Blau, 1964; Homans, 1974; Emerson, 1976), this study contributes to the understanding of service quality within the context of HRI. From a social exchange perspective, customers' evaluations of robotic service encounters reflect a dyadic assessment of what is received versus what is sacrificed, whereby perceived benefits/rewards and perceived risks/costs jointly shape the perceived value of engaging with the technology (Jiang and Kim, 2015b; Morosan and DeFranco, 2021).

In alignment with this theoretical lens, the study identifies three empirically derived dimensions (robotic service quality, robotic service efficiency, and robotic empathy) that map directly onto the foundational constructs of social exchange. Specifically, robotic service quality serves as an indicator of the overarching utilitarian benefits that customers expect from a service encounter (Kim et al., 2022). Robotic service efficiency reflects the utilitarian rewards associated with the minimization of interaction costs, such as time and cognitive effort, which are central to the cost-benefit calculus within social exchange frameworks (Morosan and DeFranco, 2021). Finally, robotic empathy represents the social rewards and affective benefits that facilitate the development of relational rapport and social presence, elements that are increasingly relevant as robots take on social roles in service delivery (Kim et al., 2022). Together, these dimensions reflect how customers assess the reciprocal value and relational outcomes in robotic service interactions. Consistent with prior frontline-robot literature (e.g., Belanche et al., 2021; Wirtz et al., 2018; van Doorn et al., 2016), customers' assessments of robotic services reflect a dual emphasis on functional service benefits (e.g., efficiency, consistency) and relational or socio-emotional value, making a hierarchical conceptualization particularly suitable for capturing both the overall evaluation and its key component drivers.

Extending this theoretical perspective into the context of robotic services is particularly significant. Traditionally, social exchange theory has focused on human-to-human interactions, highlighting reciprocity as a core component. Applying this framework to robotic contexts highlights the evolving nature of modern service encounters, where robots are increasingly functioning as interactive agents capable of forming reciprocal relationships with customers. Consistent with recent social exchange-based HRI research, customer responses to service robots are not limited to cost-benefit calculations but also reflect relational and psychological states, such as trust, rapport, and concerns about whether a robot can address customer-specific needs, i.e., uniqueness neglect (Kim et al., 2022). Accordingly, ROBOQUAL's empathy dimension is theoretically aligned with these relational exchange states, as empathic responsiveness should be especially relevant for building rapport and trust and for reducing perceived uniqueness neglect during service encounters.

Additionally, the nomological validation of the ROBOQUAL scale contributes further to understanding how robotic services influence customer perceptions and subsequent behaviors within hospitality contexts. By establishing its nomological validity, the study aligns this new instrument with established constructs in service quality literature, solidifying its theoretical position within HRI research. By adopting a direct measurement approach using ROBOQUAL, this research advocates for a more nuanced understanding of service quality, essential in contexts where conventional paradigms are evolving rapidly. This methodological innovation recognizes that the complexity of robotic services requires advanced measurement tools capable of capturing the distinctive characteristics and functionalities of robots.

Moreover, this study lays the groundwork for future research exploring the optimal integration of robotic services across different service settings, offering valuable insights regarding the design, implementation, and management of AI-enhanced service solutions. As emphasized by Kim et al. (2022), applying social exchange theory to HRI helps explain how HRI attributes shape customers' relational and psychological states, e.g., trust and rapport, which in turn influence downstream behavioral responses. Thus, this research not only enhances theoretical understanding but also establishes an essential foundation

for future explorations into robotic service quality, illustrating the transformative potential of AI technologies in redefining service interactions and elevating customer satisfaction.

4.2.2. Methodological implications

A key methodological contribution of the study is the development of an original robotic service quality scale, ROBOQUAL, which was created through a rigorous four-stage process, as depicted in Fig. 1. Complementing this, the scenario-based survey design ensured realistic representations of service interactions, thereby eliciting authentic participant responses (Busse et al., 2021). This methodological choice provided a more grounded evaluation of customer perceptions of robotic service quality compared to conventional survey techniques, which typically rely on standard survey questions that may result in biased or inaccurate responses to the abstraction gap (Qiu et al., 2020).

Moreover, the ROBOQUAL scale significantly enhances methodological rigor by offering a validated and standardized instrument to measure robotic service quality across diverse service contexts and robotic applications. This standardization fosters greater consistency and facilitates comparative studies, thus enabling the accumulation of robust empirical evidence and deepening the scholarly understanding of AI's influence on hospitality service quality.

Additionally, prescreening participants on a crowdsourcing platform to assess nomological validity before conducting the primary survey presents several advantages. This innovative approach minimizes selection bias, ensures a targeted participant sample for validity assessments, enhances data quality, improves efficiency, and is cost-effective. It also reduces respondent fatigue and strengthens the generalizability of findings. Consequently, this method provides a valuable template for subsequent research in robotic service quality across various settings.

4.2.3. Practical implications

The ROBOQUAL scale provides hospitality managers with a systematic framework for optimizing robotic service implementation and performance. Prior to deployment, managers can conduct baseline assessments using ROBOQUAL dimensions to align robot capabilities with customer expectations and property positioning. For instance, luxury properties like Aura at the Sphere in Las Vegas may discover low guest confidence in robotic empathy and strategically limit robot deployment to operationally focused tasks such as luggage delivery and late-night room service, while reserving emotionally complex interactions for human staff (Liu et al., 2024). The three-dimensional structure enables precise task-robot matching, whereby high task efficiency scores justify robot deployment for routine, consistency-dependent functions. In contrast, low empathy scores indicate a need for either advanced emotional recognition capabilities or continued human delivery of service recovery and special-occasion interactions.

Beyond initial deployment, ROBOQUAL serves as a continuous performance monitoring tool through periodic customer assessments. Declining service quality scores may signal maintenance issues or the need for software updates. At the same time, persistent dimension-specific weaknesses inform targeted improvements, such as enhancements to the emotional intelligence algorithm or modifications to the service script. This diagnostic capability extends to vendor selection processes, where managers can evaluate competing robotic systems against standardized ROBOQUAL benchmarks during pilot programs, ensuring technology investments address property-specific service priorities. Furthermore, the scale facilitates evidence-based staff training by identifying optimal points for human-robot collaboration. When ROBOQUAL assessments reveal that guests value robotic task efficiency but prefer human empathy, managers can develop seamless handoff protocols that leverage each agent's comparative advantage, optimizing both labor allocation and customer satisfaction. For example, robotic concierges that offer round-the-clock assistance and personalized recommendations enhance the customer experience, thereby improving overall satisfaction and strengthening the hotel's value proposition (Lei

et al., 2023).

The scale also enables market segmentation strategies aligned with brand positioning. Limited-service properties may emphasize and showcase high task efficiency scores as operational differentiators, while full-service hotels can use ROBOQUAL data to justify selective robot deployment that preserves their high-touch service reputation. Finally, ROBOQUAL provides measurable metrics for return-on-investment (ROI) calculations by correlating dimension improvements with key performance indicators (KPI) such as customer satisfaction scores, online review ratings, and operational cost reductions. Hotels can implement post-interaction micro-surveys measuring the three dimensions, with scores below established thresholds triggering immediate human staff intervention for service recovery, thereby creating a quality assurance system that combines robotic efficiency with human oversight.

4.3. Limitations and future research

By developing and validating the ROBOQUAL scale, this study offers valuable contributions to the understanding of HRI in the hospitality industry. Nevertheless, future research should recognize its limitations. For example, the study's exclusive emphasis on the hotel context limits the generalizability of its findings. Subsequent research could therefore expand the application of the ROBOQUAL scale into different service settings in the hospitality and tourism industries, such as restaurants, events, resorts, theme parks, and the cruise industry. Such expansions would further validate the scale's broader applicability and its potential to evaluate how robotic service quality influences customers' behavioral intentions across various service settings.

Additionally, the data collection approach adopted in this study may not sufficiently capture the full diversity of global hotel customers, given its limitation to one country and reliance on convenience sampling. Future research should therefore employ geographically and culturally diverse samples to explore how varied customer groups perceive and interact with service robots. This approach may reveal subtle cultural nuances that shape how customers perceive the quality of robotic services. Future research can also utilize field experiments or longitudinal studies to capture authentic interactions over time more comprehensively.

As service robots increasingly integrate into customer service processes and handle sensitive customer data, it is essential to rigorously examine the ethical considerations surrounding their deployment. This approach should address concerns related to privacy, data security, transparency, and potential biases within HRI. Key considerations include the methods by which service robots gather, retain, and process customer data, as well as the presence of robust safeguards to ensure data security and privacy. Finally, the potential for bias in robotic systems should be carefully considered, recognizing that training data may unintentionally perpetuate existing prejudices.

In addition, the use of a scenario-based survey method may introduce hypothetical bias, as participants' stated perceptions and intentions may not fully reflect their responses in actual service encounters with robots. Although scenario techniques are well established in service and HRI research and allow for standardized exposure to emerging technologies, they rely on imagined rather than experienced interactions. As a result, respondents may overestimate or underestimate their evaluations of robotic service quality compared to real-world settings.

Furthermore, while ROBOQUAL captures functional dimensions of robotic service quality, it does not directly measure perceptions of physical robot characteristics such as appearance, anthropomorphism, or design aesthetics. Future research should investigate how these physical attributes serve as antecedents to the functional service quality dimensions identified in this study, and whether certain physical designs create stronger or weaker relationships between service quality perceptions and behavioral outcomes. Such research could employ

experimental designs comparing customer evaluations across robots with varying physical characteristics while holding functional capabilities constant, thereby isolating the unique contribution of physical design to overall service quality assessments.

CRedit authorship contribution statement

Dogan Gursay: Writing – review & editing, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Murat Hancer:** Writing – review & editing, Supervision, Project administration. **Tingting Zhang:** Writing – review & editing, Supervision, Project administration. **Kevin Murphy:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization. **Ahmet Hacikara:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ijhm.2026.104560](https://doi.org/10.1016/j.ijhm.2026.104560).

Data availability

Data will be made available on request.

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